ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE D · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Mechatronics paper 1 · practice d.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Drawings: award marks only for components that are drawn with the correct symbol and connected correctly. Accept other correct answers.

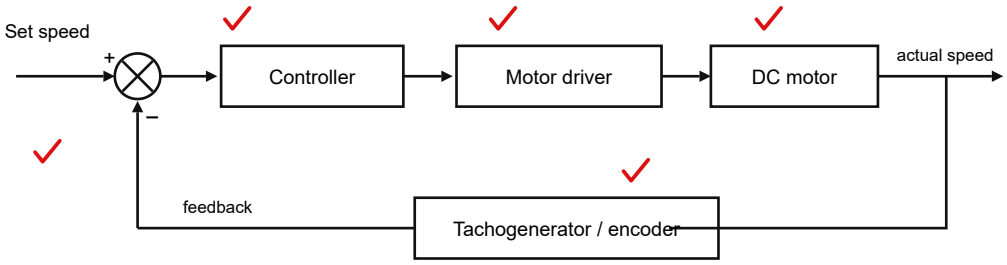
QUESTION 1

1.1 A sensor is a device that detects a physical quantity (e.g. position, temperature, light) ✓ and changes it into a signal that a controller can use ✓. (2)

1.2 Any THREE ✓✓✓: inductive proximity – metal objects; capacitive proximity – almost any material, liquids; photoelectric – objects breaking a light beam; thermocouple / PT100 – temperature; pressure transducer – pressure; encoder – shaft position or speed; ultrasonic – distance. (3)

1.3 An inductive proximity sensor detects only metal objects, using a changing magnetic field. ✓✓ (4)
A capacitive proximity sensor detects metal and non-metal objects (plastic, wood, liquids), using a changing electric field (capacitance). ✓✓ (2 × 2)

1.4



(Set speed and comparator (summing point) 1; controller 1; motor driver 1; DC motor 1; tachogenerator / encoder in the feedback path to the comparator 1) (5)

1.5 $\text{Span} = 20 - 4 = 16\ \text{mA}$ ✓ for 100 °C ✓; $\text{I} = 4 + (60 \div 100) \times 16$ ✓✓ $= 4 + 9{,}6$ ✓ $= 13{,}6\ \text{mA}$ ✓ (6)

[20]

QUESTION 2

2.1 It starts the motor smoothly by slowly increasing the voltage ✓, which reduces the high starting current and the mechanical shock on the load ✓. (2)

2.2 Any THREE ✓✓✓: the program can be changed without rewiring; fewer wires and relays, so smaller and cheaper; easier fault-finding (status LEDs, monitoring); built-in timers and counters; more reliable (no moving contacts); can communicate with other systems. (3)

2.3 When the timer coil is energised, its contacts change over only after the set time has passed ✓; they return immediately when the coil is switched off ✓. (2)

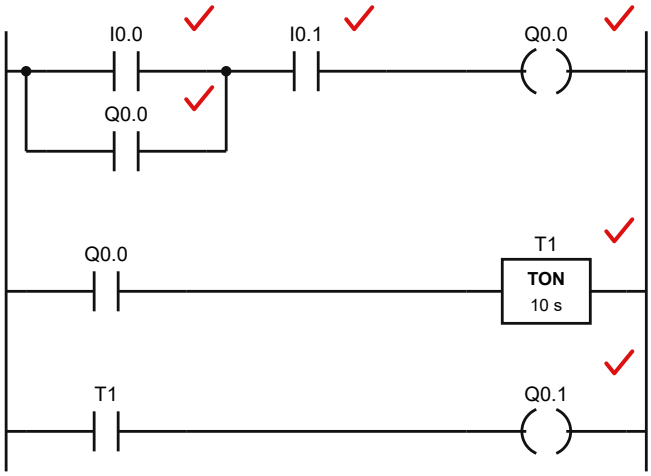
2.4.1 Stop push button (normally closed) ✓ (1)

2.4.2 Relay K1 energises ✓ and holds itself in through its own contact, so conveyor 1 runs after S1 is released ✓. (2)

2.4.3 It switches on the timer T1 ✓ when conveyor 1 starts ✓. (2)

2.4.4 10 seconds after conveyor 1 starts ✓, when the T1 contact closes and energises K2 ✓. (2)

2.5



I0.0 = S1, I0.1 = S0 (wired NC), Q0.0 = conveyor 1, Q0.1 = conveyor 2

(I0.0 NO 1 with a Q0.0 holding contact in parallel 1; I0.1 in series (programmed NO because it is wired NC) 1; output Q0.0 1; Q0.0 starting timer T1 set to 10 s 1; T1 contact driving Q0.1 1)

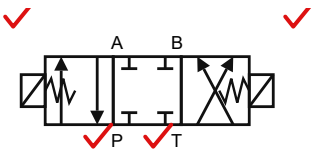
(6)

[20]

QUESTION 3

3.1 Any THREE ✓✓✓: gear pump; vane pump; piston pump; screw pump. Gear pump advantage, any ONE ✓: simple and cheap; robust; tolerates dirt better; easy to maintain. (4)

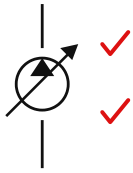
3.2.1



(Three squares 1 with the closed centre (all ports blocked) 1; solenoids on both sides 1; centring springs on both sides 1)

(4)

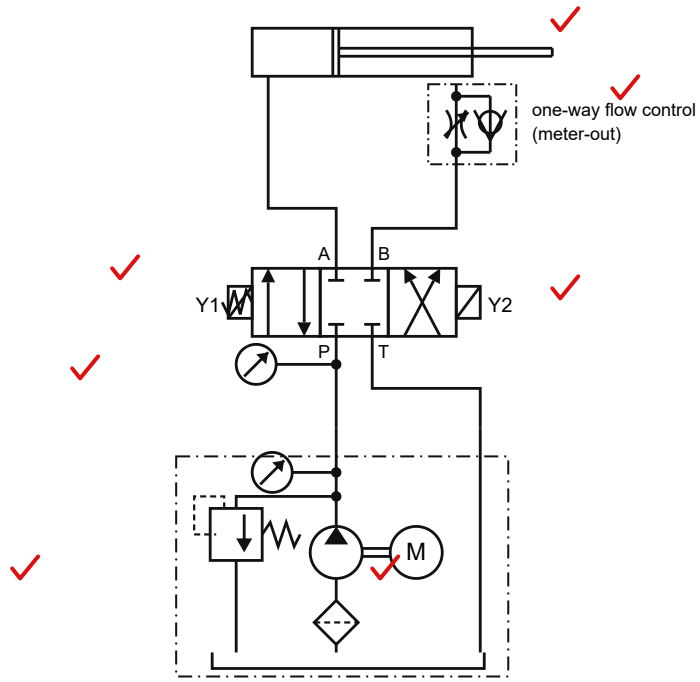
3.2.2



(Pump circle with ONE triangle pointing outwards 1; diagonal arrow showing that it is variable 1)

(2)

3.3



(Power pack with relief valve 2; pressure gauge 1; 4/3 double-solenoid, spring-centred, closed-centre valve 2; double-acting cylinder correctly connected 2; one-way flow control in the rod-end line, with the check valve letting oil in freely, so the oil leaving during extension is throttled (meter-out) 3)

(10)

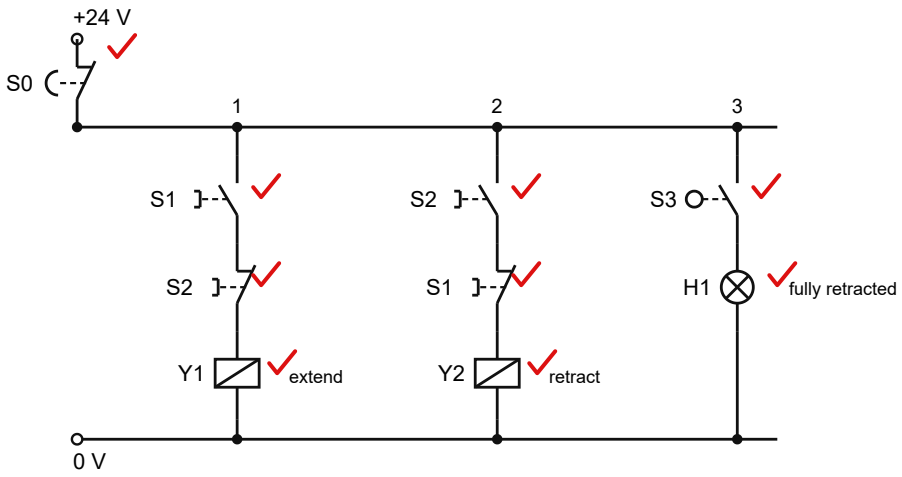
[20]

QUESTION 4

4.1 It detects the position of the cylinder or workpiece without touching it ✓ and sends a signal to the control circuit (e.g. to stop the stroke or start the next step) ✓. (2)

4.2 Any FOUR ✓✓✓✓: emergency stop button; fuses / circuit breakers; overload relays; guard (safety) switches; two-hand control; pressure switches; safety relays; light curtains. (4)

4.3



The NC contact of each push button opens the other circuit, so Y1 and Y2 can never be on together.

(Supply lines 1; E-stop (NC) in the supply 1; S1 (NO) 1 in series with the NC contact of S2 1 driving Y1 1; S2 (NO) 1 in series with the NC contact of S1 1 driving Y2 1; no holding contacts, so the cylinder moves only while a button is held 2; S3 (NO, actuated at full retraction) switching lamp H1 2; correct symbols 1; neat 1)

(14)

[20]

QUESTION 5

5.1 In order ✓✓✓✓✓: collect information about the fault (ask the operator, look at the symptoms); analyse the information and study the circuit diagrams; locate the faulty part (test and measure, e.g. half-split); repair or replace the faulty part; test the system and record the fault and repair. (5)

5.2 Any FOUR ✓✓✓✓: no voltage reaching the coil (blown fuse, broken wire, faulty contact); burnt-out (open-circuit) coil; spool stuck because of dirty oil or varnish; wrong voltage on the coil; low pilot pressure (pilot-operated valve); mechanical damage / spring broken. (4)

5.3 Test at a point about halfway along the circuit or system ✓. If the signal is correct there, the fault is in the second half; if not, it is in the first half ✓. Keep halving the faulty part ✓ until the faulty component is found – this saves time ✓. (4)

5.4 Any FOUR ✓✓✓✓: leaks at the fittings; cracks or bulges in the outer cover; kinks or twisting; rubbing or wear against other parts; correct pressure rating; age / date on the hose; loose or damaged fittings. (4)

5.5 It records all maintenance, faults and repairs ✓, which helps to plan maintenance and spot repeating faults ✓, and shows that legal safety inspections were done ✓. (3)

[20]

TOTAL: 100

END OF MARKING GUIDELINE